## 4.4 Multiplier

The concept of multiplier was first developed by R.F. Khan in terms of employment. J.M Keynes redefined it as investment multiplier.

The multiplier is defined as the ratio of the change in national income to change in investment. If ΔI stands for increase in investment and ΔY stands for resultant increase in income, the multiplier K =ΔY/ΔI. Since ΔY results from ΔI, the multiplier is called investment multiplier.

### 4.4.1 Assumptions of Multiplier:

Keynes's theory of the multiplier works under certain assumptions which limit the operation of the multiplier. They are as follows:

1. There is change in autonomous investment.
2. There is no induced investment
3. The marginal propensity to consume is constant.
4. Consumption is a function of current income.
5. There are no time lags in the multiplier process.
6. Consumer goods are available in response to effective demand for them.
7. There is a closed economy unaffected by foreign influences.
8. There are no changes in prices.
9. There is less than full employment level in the economy.

### 4.4.2. Marginal propensity to consume and multiplier.

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The propensity to consume refers to the portion of income spent on consumption. The MPC refers to the relation between change in consumption (C) and change in income(Y).

Symbolically $MPC = \Delta C/\Delta Y$

The value of multiplier depends on MPC

Multiplier $(K) = 1/1\text{-}MPC$

The multiplier is the reciprocal of one minus marginal propensity to consume. Since marginal propensity to save is 1 - MPC. (MPC+MPS =1). Multiplier is 1/ MPS. The multiplier is therefore defined as reciprocal of MPS. Multiplier is inversely related to MPS and directly with MPC.

Numerically if MPC is 0.75, MPS is 0.25 and k is 4.

Using formula k = 1/1- MPC

1/1-0.75 =1/0.25 =4

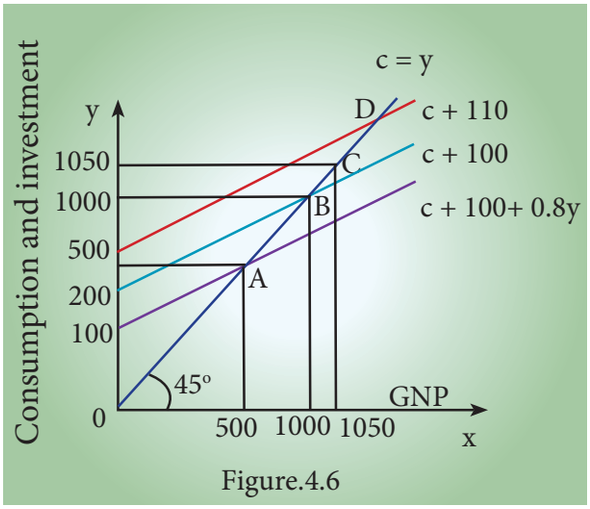



Figure.4.6



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